

SAI TEJA ADEPU

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Portfolio: portfolio-psi-ten-5kxb2gvn1q.vercel.app

PROFESSIONAL SUMMARY

AI Engineer specialized in Generative AI, Agentic Workflows, and Machine Learning. Expertise in architecting end-to-end RAG pipelines and stateful AI agents using LangChain and LangGraph. Highly proficient in Python-driven automation, vector database optimization (ChromaDB), and deploying scalable backend services via FastAPI. Focused on developing production-grade AI solutions with superior retrieval accuracy and LLM orchestration.

TECHNICAL SKILLS

- **Languages:** Python, SQL
- **Agentic AI & GenAI:** LangGraph (Stateful Agents), LangChain, LangSmith (Monitoring), RAG pipelines, Prompt Engineering, Semantic Search, Embeddings
- **Machine Learning & Deep Learning:** Model Training/Optimization, NLP, TensorFlow, Keras, Scikit-learn
- **Backend & Data Infrastructure:** FastAPI, ChromaDB, Streamlit, API Integrations
- **DevOps & Tools:** Git, GitHub, VS Code, Google Colab, Linux/Unix

KEY AI PROJECTS

Conversational RAG with PDF Upload & Chat History *LangGraph, LangChain, ChromaDB, Groq*

- Engineered a stateful multi-turn RAG application using LangGraph to manage complex conversation flows and document querying.
- Implemented history-aware retrievers for contextual reformulation, ensuring high precision in multi-turn enterprise support scenarios.
- Optimized document ingestion pipelines using RecursiveCharacterTextSplitter and PyPDFLoader for efficient vector storage.

PDF Question Answering System (RAG-based)

LangChain, ChromaDB, Ollama

- Built a robust retrieval-based QA system with integrated performance monitoring for query execution latency and response quality.
- Managed high-dimensional vector search spaces to fetch highly relevant context, significantly reducing LLM hallucinations.

Agentic LLM Chatbot with Configurable Parameters

LangChain, Ollama, Streamlit

- Developed an interactive AI agent allowing dynamic user control over hyper-parameters (temperature, max tokens) for tailored outputs.
- Utilized advanced prompt templates and structured output parsing to ensure consistent response formatting for downstream tasks.

Advanced Image Classification (Dog vs Cat)

TensorFlow, Keras

- Built and deployed a CNN model optimized through data augmentation and preprocessing to achieve high classification accuracy.

EDUCATION

M.Sc. Data Science / Computer Science	2024 – 2026
Osmania University	CGPA: 8.01
B.Sc. Computer Science / Data Science	2020 – 2023
Telangana University	CGPA: 8.01

CERTIFICATIONS

- **Generative AI with LangChain & Hugging Face** – Udemy
- **Python for Data Science** – Simplilearn
- **Machine Learning** – Simplilearn